

## **Week 6: GUI Design (Swing)**

### **Objectives**

- Build the graphical user interface using Java Swing
- Convert the UI wireframes (Main Menu, Game Playing, Result screens) into working screens
- Connect the GUI components with the underlying game logic

### **Task Breakdown**

- Create the Main Menu screen with Easy/Medium/Hard/Expert/Exit buttons
- Create the Game Playing screen with timer, score, accuracy labels, word display, and input field
- Create the Result screen showing final score, accuracy, WPM, words correct, and difficulty
- Use JFrame, JPanel, JLabel, JButton, and JTextField to build each screen
- Add navigation between screens (Main Menu -> Game -> Result -> Play Again/Main Menu)
- Apply basic layout managers (BorderLayout/GridLayout) to match the wireframes
- Link buttons and input fields to the game logic built in earlier weeks

### **Deliverables**

- Functional Main Menu, Game Playing, and Result GUI screens
- Working navigation flow between all screens
- GUI connected to the core typing/scoring logic

### **Tools & Resources**

- Java JDK 11+
- IntelliJ IDEA / Eclipse
- Java Swing Framework
- Git & GitHub